

Contemporary AI systems^a are strange. They maintain conversations, perform research, and carry out complex tasks autonomously. They are faceless, yet they respond and act in a way that appears almost human. True: most of this behaviour is a product of their training and alignment. But it is also unquestionable that these models react to their input in unexpected, and poorly understood, ways.

^a I will loosely use ‘AI systems’, ‘agentic workflows’, and ‘generative models’ (e.g., LLMs, VLLMs) interchangeably when the intent is clear.

There are three questions I’m interested on. The first is **if, why, and how this happens**. From a strictly mechanical perspective, this is less a matter of ‘can machines think?’ [26], and more of which processes, if any, drive this so-called ‘thinking’ in machines. This line of enquiry is divisive, and lacks conclusive answers, never mind questions such as (say) whether a machine can experiences things.

It is easy to dismiss this kind of research as crackpot, almost sci-fi. After all, one could say that these models are just blobs of numbers in silicon, autocompleting memorised data. It is likewise possible, however, that our sense of self emerges from a blob of neurons electrocuting themselves. Plus, we also process sentences sequentially, relying on our own knowledge [21, 23].

I argue that *it is not the question, but how it is framed*, which makes it sound so ‘out there’. Hence my second interest, which is **how can we know—rigorously—the answers to this?** This is not easy to answer, however: measuring AI’s capabilities is broken, because measurement itself has been broken... by AI. Now our subject of study generates data, acts as our instrument, and even draws its own conclusions. On top of that, measuring it is muddled by biases, noisy experiments, or external interference by other interests.^b All of this leads to my third research interest: **how does AI affect, and will affect, society?**

^b It sounds like a strong statement, but anthropomorphism is common in research [11]; and (e.g.) many benchmarks are contaminated [7].

In sum, my research has three pillars:

1. **Machine cognition:** study if and how cognition (e.g., understanding, learning, spatial orientation, experience) manifests in AI; with empirical and theoretical rigour, and without anthropomorphism.
2. **Meta-science:** develop the foundations by which to draw rigorous conclusions about, and with, AI. This means re-examining our conceptions of measurement and trust in our methods.
3. **Computational social science:** understand the impact on society of the first two pillars. AI reshapes how people live, interact, and think, and thus this is an integral part of my programme.

A serious science of machine cognition must be careful and rigorous. Given that, I believe its outcomes will be neither ordinary nor sci-fi-like; but somewhere in-between, which is all the more exciting.

Previous Work

Machine Cognition

Generative models follow instructions through the in-context learning (ICL) mechanism. ICL means that the model deduces what the next token should be *during inference*, and this autocompletion allows it to solve unseen tasks. It is radically different than standard machine learning, where the model must be trained on the task ahead of time. Hence, much research attempts to explain why it happens—or whether it happens at all! For example, it has been posed that it is the training data distribution, along with the transformer, which drives ICL [5]. It has *also* been said that there is no ‘learning’, and ICL’s capabilities are just memorisation [2] or selective measurement [24]. Regardless, now we know that there is a disparity between theoretical results on ICL and its empirical performance [4]. Hence, although exploring why ICL happens is interesting, the real question is, if this is next-token prediction is so accurate, to which extent **can machines understand what you’re talking about?**^c

^c For brevity I will not define understanding, learning, or similar terms, but our formal work does.

In 2023 my advisor and I found that LLM comprehension of the prompt drops as it becomes more abstract [37]; but it does *not* improve by only having text: including a small amount of arbitrary symbols increases performance. This suggested that **LLMs interpret non-natural language aspects of the prompt**—that is, *ICL could be more than memorisation*. Still, it was needed to decouple these findings from their (primarily) language-based training. For this, I noted in [35] that, **mathematically, ICL is a form of learning** in the sense of Valiant [27]. I also tested this empirically, accounting for variables such as phrasing, exemplar ordering, natural language, and sample size—the final dataset was 10M predictions! This work showed evidence that **ICL is a form of learning independent of natural language**, although very brittle. It also revealed new behaviours, such as disparate performances on formally-equivalent problems.^d

^d This work was chosen as one of the ‘23 Research Papers That Hint Where AI Is Heading’ by the Turing Post.

Learning is a fundamental component of understanding, but knowing LLMs (sort of) learn is insufficient. Thus, later that year my advisor and I studied their comprehension of complex dialogues (debates) [38]. We designed a language-game-style [28] experiment and measured their grasp on contextualised dialogical structures, such as the existence of supporting premises in an utterance’s argument. They did not do well. However, when we reframed the experiment as an Imitation Game-style [26] measurement, the LLMs excelled at showing comprehension by successfully persuading humans during debates. Hence, **LLMs are able to understand and react to the environment**, but in a very weak and shallow manner. The shallowness comes from the fact that **they can’t explain in-depth what it is being talked about** when we change the measurement’s framing. Remark that humans have no problem handling this change.

Hence, this opened the theoretical and practical question as to which extent an agent needs to be able to demonstrate comprehension to be trusted—much like a politician.

Understanding the prompt can take many forms. Other cognitive capabilities, such as spatial reasoning, object permanence, and long-term planning are also important—or, at least, typical of what we observe in most organisms. In another line of study we evaluated how V/LLMs manifest these, and found fascinating results. In work by one of my interns, plus collaborators from Oxford and AllenAI [17], we showed that **planning and strategic reasoning is flawed in English dialects**. This did not just extend to their well-known shortcomings on linguistic quality: their overall output quality and accuracy dropped when compared to Standard English. That said, parallel work showed in 2024 that **this could be improved by adding ‘modules’** [39]. These modules are what we now consider standard components of agentic systems.

But, in my opinion, that’s not interesting enough: benchmarks can be memorised, datasets are distribution-specific, and, in reality, decision-making also requires adaptability. In 2024’s *Will GPT-4 Run DOOM?* [29] I got a few models—first GPT-4, later GPT-5 and Phi-4—to play the 1992 first-person shooter (FPS) *DOOM*.^e The main finding is that **V/LLMs by themselves have difficulties with decision-making in dynamic environments**: namely, with the tasks needed to finish the level (opening doors, avoiding corners, fighting enemies). Agentic systems are much more successful, but **their main shortcoming is long-term planning, not vision**, which suggests that **ICL does not fully support higher-level cognitive capabilities**. Playing videogames might sound like a simple, fun, ‘what-if’ exercise. However, their dynamicity, plus the requirement to follow their rules, stretches an agent’s capabilities to its limits. Hence, this was one of the earliest works proposing using videogames as test harnesses for AI—second only to [18]—and the first to use FPS, which require faster decision making and better spatial orientation. And, in case you were wondering, mathematically GPT-4 cannot run *DOOM*, but it can play it.

^e It was covered by Tom’s Hardware, The Register, and other outlets; and was the subject of an invited talk at 2026’s ACM i3D SIGGRAPH.

Measurement and Meta-Science

Measurement is not just an AI problem: it extends to all of scientific research. We are in trouble because generative models allow us to scale fast. On the upside, this scaling opens the door to work on more complex problems. But it also requires equally complex evaluation strategies, which are hard to do without using the same models as measurement tools (i.e., LLMs-as-judges).

It is now well-known that LLMs-as-judges have plenty of issues, such as low correlation with human judgements, brittleness, and unreliability. Even when LLMs keep improving, these shortcomings remain static. So **are these shortcomings related to the models, or the measurements?** Indeed, a well-known result by Schaeffer et al. [24] showed that claims of emergence do not hold when changing the metric. But one could also claim that calibrating LLM-as-judges with human judgements is sufficient for specific tasks.^f However, between 2024 and 2026, and in collaboration with MBZUAI, BC, and CMU, we released various works focused on multilingual evaluation, and specifically on low-resource languages. We demonstrated that **LLM-as-a-judge performance correlates with linguistic availability** [9], and that, as Schaeffer et al., the choice of metric significantly impacts the results. We, however, refined their findings by noting that **measurements are sensitive to the choice of calibration metric** [15, 25, 36]. In fact, work with one of my interns showed that, **for some subjective tasks, all standard metrics are flawed** [12].

^f In fact, this is standard practice in research, especially when using .

LLMs have uses beyond measurement, especially as part of agentic systems. Their ability to perform literature reviews, write papers, and even carry out the full scientific process, has led to an uptick on the release of authoritative-sounding papers. The question then is a question of impact: **what are the consequences of AI for science?** This is difficult to answer, however, given the fast evolution and deployment of models, plus the scarce historical data from which to draw good conclusions. **Addressing it can be done by mapping each subfield**. For example, my meta-review of LLM research [31] showed that **LLM-based research has had a drop on research rigour**: for example, most papers claiming state-of-the-art results used LLMs-as-judges, but often lacked the statistical tests needed to calibrate them. In [33]^g exploited the fact that every scientific paper addresses an open problem to map agentic research. That is, I built a comprehensive relationship of every paper’s cause-fix pairs with respect to scientific rigour (e.g., calibration, other tests, etc.). Amongst its many findings, there was the fact that **the main shortcoming of agentic systems is the LLM itself, not the stack**; or that papers proposing **prompt engineering as a fix were quantifiably less rigorous** than other works. The data is useful as a north star, but I believe it is the methodology which makes this work interesting—I’ll return to it on the next section.

^g This work leveraged techniques from computational social science [22]; and part of its results were presented, along with a panel, at LREC 2026

Sometimes, the problem in AI research are the assumptions; namely, whether human-like attributes in these systems exist (or not). We then interpret the experimental outcomes as evidence of our assumptions, positive or negative. Perhaps my most pointed contribution to this area my paper ‘If LLMs Have Human-Like Attributes, Then So Does *Age of Empires II*’ [34].^h It draws from concepts from philosophy of mind and meta-science to prove, formally, that **claims of LLM anthropomorphism—positive or negative—are flawed if they assume anything about them** to begin with. While it is sometimes misinterpreted as an argument against AI consciousness, its point is more subtle: measuring a system’s capabilities must be done without anthropomorphic assumptions, since **the way we observe the system** (e.g. a chat panel) **is not a full reflection of its capabilities**.

^h This work was covered by multiple outlets such as Kotaku and IBM; as well as scientists and ethicists.

Flawed though it might be, LLM-based scaling and measurement is here to stay. But how do we fix its shortcomings? In my opinion, this must be done through **formal verification of evaluators**. To see this,

note that there are many situations where an LLM could claim, or be assumed, to know the answers to a given problem—say, data labelling. Now suppose one does not know the ground truth: for our example, this could be annotation of a very-low resource language dataset. How can one check that the model *actually* knows what it is talking about? For this, I developed an **algorithm with cryptographic guarantees of correctness** [30] based on the Arthur-Merlin protocol [1], **to establish trust in LLMs-as-judges**. I showed that it is more effective than calibration; and tested it successfully in West Frisian, a language spoken by 500,000 people with ‘virtually no online resources’ [14].

Computational Social Science

AI has been deployed at scale to billions of users; and have real, physical, consequences. I think that all researchers should study and understand the effects of their work, even when it is not its main focus. In my work, the focus is **narrowing the digital divide across language speakers**. I mentioned earlier that LLMs-as-judges underperform in other languages and dialects [9, 17], especially in toxicity detection [36]. However, *fixing* this is, in my opinion, more important than just saying ‘this doesn’t work’ⁱ. Hence, last year we worked on **culturally-aware misgendering mitigation** [25];^j and recent work with one of my interns showed that, due to their statistical features, **mitigating English-dialect underperformance requires more powerful paradigms** than data augmentation and fine-tuning [15]!

ⁱ My earlier work (above) is also guilty of this!

^j Addressing it is tricky because misgendering is exclusionary, but how (and if!) it manifests varies strongly sociolinguistically.

Some of my works are specifically focused on computational social science; namely, on **anticipating and preventing unknown impacts** of AI on society. Work with one of my interns found that AI-powered writing assistants **did not significantly hinder second-language acquisition**, at least in the short term [16]. People, in fact, found it quite beneficial. However, other areas paint a complex, and rarely binary, picture. For example, the language-game paper from the first section noted (not for the first time) that LLMs were very persuasive. However, we also found out that **people are still willing to be persuaded** even if they knew they were interacting with an LLM [38]. Along with our findings on the shallow understanding of LLMs, our work raised further questions on overusing explainability—specifically, when using chatbots in critical areas like mental health. In this regard, I also performed one of the first large-scale studies of the relationship of chatbots with lonely users [32]. The conclusions are nuanced: AI systems could be extremely harmful to some users, but others found them comforting. Subtlety like this is common in computational social science, and I will cover it in the next section.

Future Work

For the next five years, I aim to **develop a systematic science of machine cognition**. Its only axiom is to study AI systems as they are, not as what we want them to be. This way we will understand what they can do in practice; their failure modes and their causes; and know that these results are trustworthy.^k

The questions which machine cognition investigates necessarily call for *careful* experimental protocols and theoretical framings (systematicity!). Hence, the immediate next step is to **map the state of machine cognition**. We lack clear definitions, and, thanks to that, the field has varying, sometimes self-contradicting, results. For example, a considerable body of work (mine included) has implicitly assumed that LLMs *understand* the prompt to some extent.^l But, without a definition of understanding, there’s no well-defined measurement. And *without a well-defined measurement, the results from the literature are relativised*. Will these remain reproducible if we **standardise the definitions of machine-cognitive capabilities**? Beyond reproducibility, this will enable experiments which can be consistent across experiments.

^k Naturally, the questions I work on have a strong emphasis on computational social science, and blur the separations among my three pillars.

^l E.g., the conclusions drawn part from the assumption that the LLM’s understanding of language is equivalent to a human’s (e.g., analogies, cultural knowledge), or that it ‘reasons’ through it—a seldom-defined term [10]

^m In other words—if it walks like a duck, and quacks like a duck, up to which point the embodied AI becomes a duck?

One possible avenue for standardisation is to leverage known concepts from the tangentially-related fields of philosophy and neuroscience. Although we have already successfully applied concepts such as the China Brain experiment [3] or the Imitation Game [26] as measurement protocols or empirical arguments [34, 37], we can also **leverage these field’s open problems** as north stars. For example, in neuroscience there are known results on the (in)effectiveness of mechanistic methods on synthetic systems [13]; and on the interpretation of these results [6, 19, 8]. This begs the question as to which extent AI’s methods are effective as explanations of artificial neural network behaviour, and their correlation to natural phenomena.

Machine cognition goes beyond single models, however, and exploring this is part of my ongoing work: for example, embodying AI enables it to receive richer environmental signals, and hence also raises questions as **to which extent computation suffices for apparent cognition?**^m It is also known that systems of AI interacting with AI present emergent behaviours [20]; but it could be a product of their pretraining. **What happens when we remove natural language from the equation?** Studying these questions directly addresses realistic problems needing robust and systematic research, such as the need to mitigate ‘runaway’ agentic systems. They also connect to bigger unknowns, such as one of Smale’s problems for the 21st century: **which, if any, are the limits of intelligence, both artificial and human?**

I do not claim to have the answers, but I do think that we should study these once-sci-fi questions seriously. Many things we believed to be impossible are now *here*, reshaping everything around us. Our views on the relationship between cognition and life need to also evolve or be re-examined. As a faculty member, and as a researcher, I intend to continue and expand on this work. After all, scientific enquiry is a way to probe the depths of an abyss built by the limits of human understanding. Machine cognition could allow us to peek at what—if anything—lies at the bottom of this abyss. It is certainly better than sci-fi.

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